

Summary for Assignment 4

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Consider the IMDB example from Chapter 6. Re-run the example modifying the following:

1. Cutoff reviews after 150 words.
2. Restrict training samples to 100.
3. Validate on 10,000 samples.
4. Consider only the top 10,000 words.
5. Consider both a embedding layer, and a pretrained word embedding. Which approach did better? Now try changing the number of training samples to determine at what point the embedding layer gives better performance.

Considerations 1-4 are included in my r markdown file. Consideration 5 is answered below:

With only 100 training samples, the pretrained GloVe embedding clearly performed better than the learned embedding, which basically stayed at random guessing (around 0.50 accuracy) while GloVe reached about 0.54. This makes sense since the learned embedding doesn't have enough data to figure out meaningful word relationships, while GloVe already has that built in from a much larger dataset. When I increased the training size, the learned embedding started to improve a lot. GloVe still did better at 200 samples, but by around 500 samples the learned embedding caught up and actually started outperforming it. From there, the learned model continued to improve and stayed ahead at 1000 and 2000 samples. So overall, GloVe works better when you have very little data, but once you have enough training examples, the learned embedding ends up being the stronger approach.

Learned embedding:

```
learned_eval <- learned_model %>% evaluate(test_ds, verbose = 0)
learned_eval
```

```
##      loss  accuracy
## 0.6935223 0.5000000
```

Pretrained GloVe embedding:

```
pretrained_eval <- pretrained_model %>% evaluate(test_ds, verbose = 0)
pretrained_eval
```

```
##      loss  accuracy
## 0.7011006 0.5444400
```

Learned embedding became better at 500 training samples as the table below states:

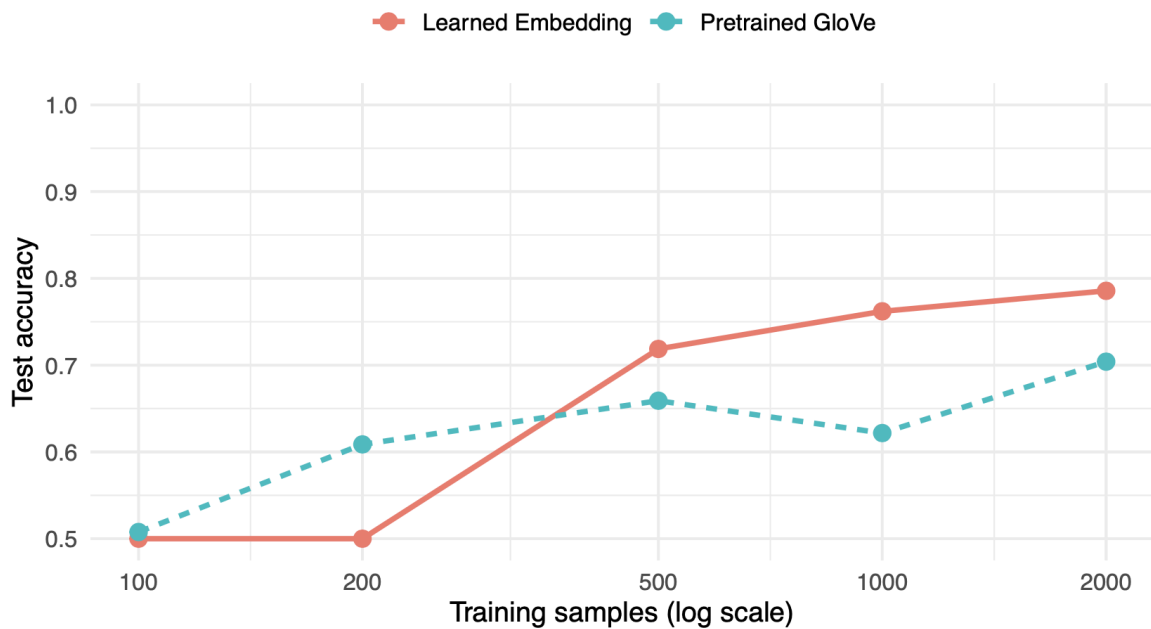
Table 2: Accuracy vs. Training Size

Training Samples	Learned Embedding	Pretrained GloVe
100	0.5000	0.5077
200	0.5000	0.6087
500	0.7187	0.6590
1000	0.7620	0.6217
2000	0.7857	0.7040

I started by moving up from 100 to 200 training samples and the learned embedding didn't move, but the pretrained did get better. Once 500 training samples was introduced, learned shot up and had better accuracy. This trend continued with 1,000 and 2,000 samples as well.

Accuracy vs. Training Sample Size

Learned vs. Pretrained GloVe



Conclusion:

As the number of training samples increases, the learned embedding improves.